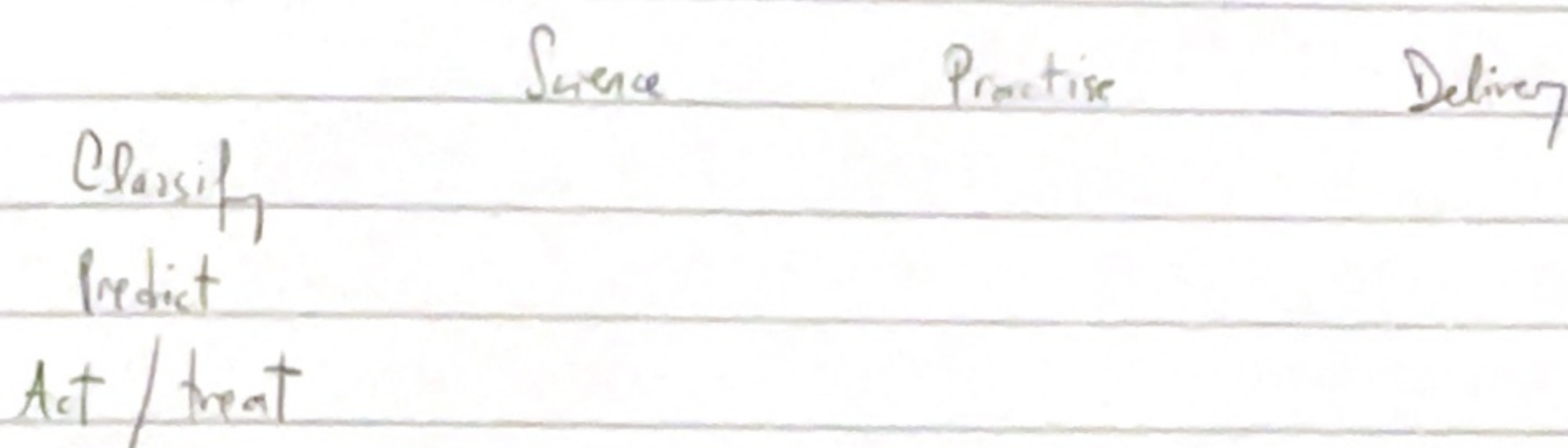


⑥

Designing and evaluating clinical ML applications

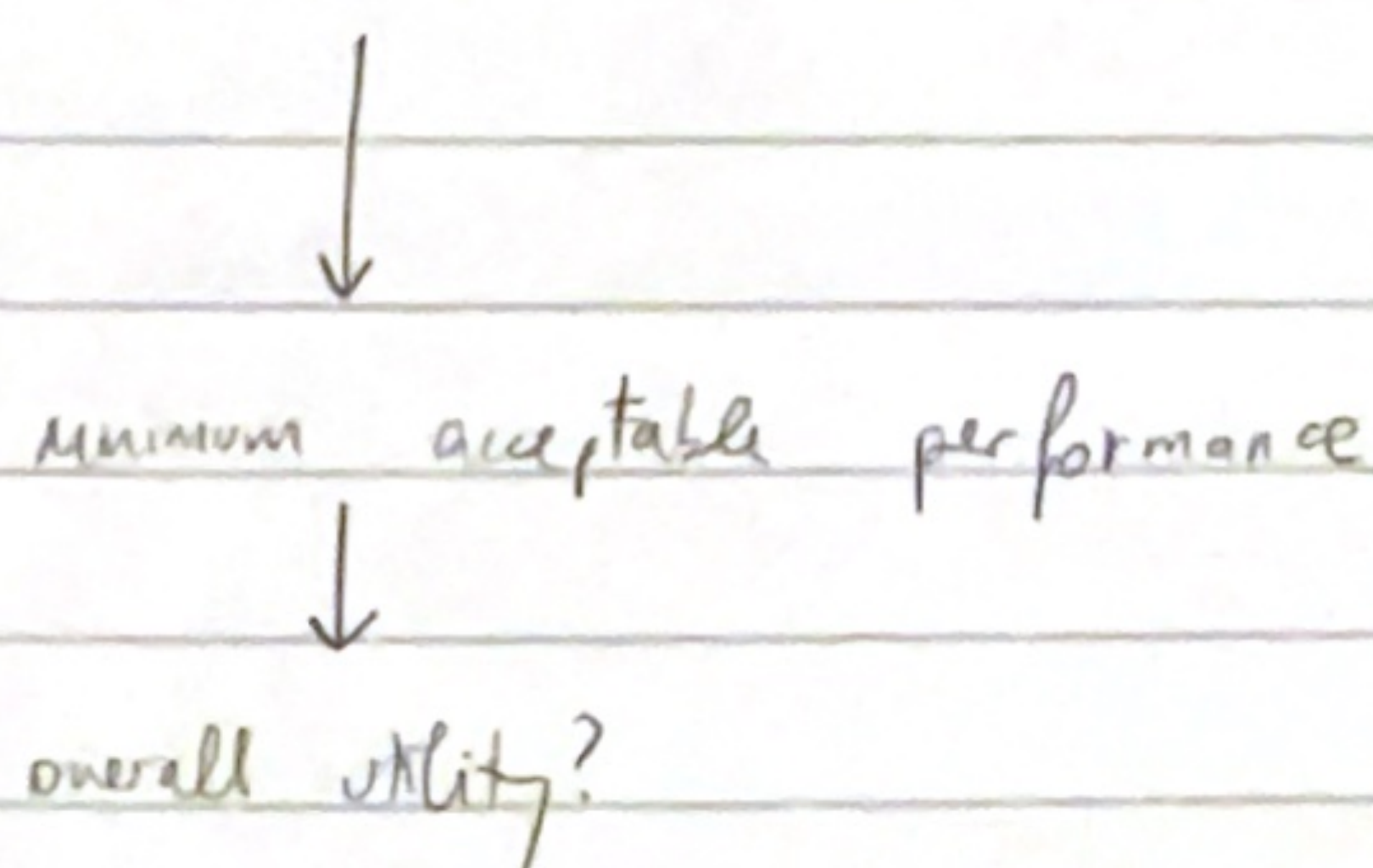
→ finding problems worth solving



⇒ begin with the end in mind

⇒ OAP = Output Action pairing

⇒ OAP framework



(slides) ⇒ ≠ factors to consider

Computer Scientist /
engineer

Data Scientist

ML techniques
co-develop model
workflow pipeline
coding
↓

knows ML techniques

data mining
feature engineering → ge processing
etc
analytics
metrics of model performance
↓
can do simple models

← agency can optimise this

FOUNDATION MODELS

(FM)

def → learned from ^{unlabelled data} unsupervised

adaptability → learn ↑ → with ↑ data

↑ n° of parameters

TRANSFER LEARNING

create vision of world from ↑ data
self supervised

fine tuning
→ these tweaks

few shot

zero shot

chef that
is asked

new recipe

FM adapts to
new tasks with
small training

eg. skin lesions

now wisdom

FM transfers
knowledge to
new tasks

eg. likelihood
of a pt developing
a condition

(or rare conditions)

Other (+)

→ enabling

models models
reusable
robust

→ open new AI interfaces

→ useful where data ← scarce
diverse

Narrow AI

1 touch

programmed by human
Doesn't "emerge"

General AI

many tanks

learns and teaches itself

emergent behaviours

this system $\left\{ \begin{array}{l} \uparrow \text{data} \end{array} \right.$ $\xleftrightarrow{\text{interrelation ship}}$ $\uparrow \text{parameters at scale}$

exhibits behaviours not predictable from data alone

↓
et. → flock → collective behaviour consequence of relations between individual elements

👁 hallucinations → generation of false output

unrealistic

↓ FM should be trained with data
to ↓ its incidence

predictions validated through rigorous testing and experimentation

develop govt governing capabilities - of AI to check AI

How do FM work

key components → transformer architecture → 1st introduced in paper
"attention is all you need"

Neural Model



→ Attention mechanism

→ Self-attention : input data ^{applies} → attention mechanism → itself

→ Considers → relations ^{between} → elements → All at once

LLM



high level = Deep Neural Network

⇒ Core = layers interleaved w/ attention mechanisms

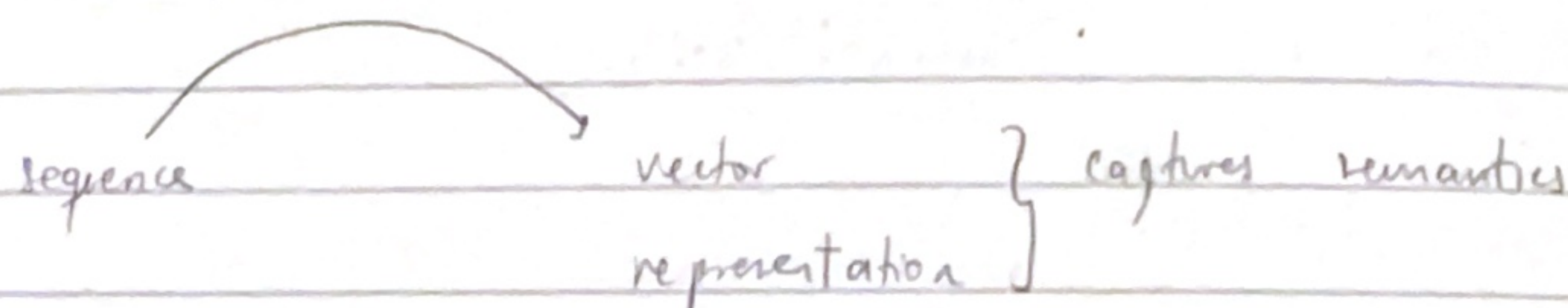
⇒ Sequential data

⇒ Tokenisation ^{breaks} → text to units

→ can be applied to other things ≠ than text

⇒ Subtypes of transformer models

→ transformer encoder



Self supervised
training
with large
data sets

unlabeled data

eg BERT — there are unlabeled
words → model
tries to find
them and predict

then pre-trained w/ small amounts of labeled data

then
(trained w/)
GRADIENT
DESCENT

⇒ Model that generates its own labels based on inherent structure
is that unlabeled data instead of requiring human labels

→ transformer decoder



eg ChatGPT

Self supervised

Pre-trained

→ predicts words given part words

Objective functions:

→ Probability distribution over word sequences

Chat GPT based on GPT3 foundation LLM

↓
Reinforced learning (trained receiving human feedback)

3 key steps

→ supervised fine tuning
→ mimic human preferences → Reward model
→ proximal policy optimization

→ PROMPT = input model uses
starting point

control what generates
direct attention

→ Prompt engineering → select / → data → to train a model
constructing /
organizing /
setting
constraints

goal → maximize accuracy / efficiency
↓ hallucinations

Types:

→ Instruction prompt → we ask a task

→ Role assigned prompt → we assign the model a role

→ few shot standard prompt → includes examples of the task
the prompt is trying to solve

→ Chain of thought prompt → model explains reasoning

⇒ zero shot chain of thought prompt → when you don't have an idea of the answer

→ Self consistency prompting → ^{aim at} improving the accuracy of the model's response

↓
By Generating multiple chains of thought +

taking the majority answer as the final answer

↓
Could improve =

DA / PT management / T10s planning

Generative knowledge → Generation → facts

↓
Integration → based on the generated facts and then chooses the + probable

→ MULTIMODAL Models → DALLI family

→ DIFFUSION MODELS → generates realistic images

→ Tokenisation

- Named entity recognition → drugs / diseases
- negation detection → T/F on entities
- relation extraction → groups of phrases together
- de-identification → protect privacy + confidentiality

} require purpose
built-
solutions
|

⇒ FM could be trained with ↑ data to
↑ accuracy

FM can do
it easier
|

Also generalisation

⇒ Genomic → FASTA format